tcsBU - a TCS beautifier

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Introduction

TCS (Clement *et al.*, 2000) is among the most popular haplotype network reconstruction methods. Written in Java, it is based on the *Statistical Parsimony* algorithm developed by Templeton *et al.* (1992). Unfortunately, TCS has some major drawbacks related with its output: 1) the graphics are coarse and often not ready for publication; 2) there is no built-in way of classifying haplotypes and displaying that information in haplogroups; and 3) when more than one network is present in the data, the software cannot use its 'spring' algorithm to layout multiple graphs, resulting in overlapping networks that have to be disentangled manually.

tcsBU (read as *TCS Bee You*) was developed to allow the production of publication-ready networks resulting from TCS analysis without too much effort. Users are able to:

- Classify haplotypes (for example, according to sampling locations or dates) and display this information as pie-chart-like haplogroups within the network.
- Load continuous trait data (such as collection time or any numeric variable) and visualize it as an inner ring within each node, using a color gradient.
- Display haplotype names (e.g., H1, H2) on nodes when a three-column haplotype configuration file is provided.
- Save the final network as a Scalable Vector Graphics (SVG) file for publication or further editing.

The software

Dependencies

tcsBU is a browser-based JavaScript program that can be served by an HTTP server or run directly from a local folder. All computations are performed by the browser. **tcsBU** depends on the following third-party libraries (license type in parenthesis):

- JQuery (MIT): the underlying Javascript engine
- w2ui (MIT): JQuery-based User Interface
- d3.js (BSD): graphics engine (force-directed graph layout)
- FileSaver.js (X11/MIT): implements file saving in the browser
- d3-svg-legend (MIT): a modified version integrated into **tcsBU** source code

tcsBU will only run on modern browsers (Firefox 20+, Chrome, Chrome for Android, IE 10+, Opera 15+, and Safari 6.1+).

Installation

For a web-based installation, save the **tcsBU** package into any directory under the webserver ROOT directory. No further steps are necessary. The only files/folders required are:

- index.html
- help.html
- tcsBU.js
- tcsBU.min.css
- jquery-3.2.1.min.js
- w2ui.min.js / w2ui.min.css
- d3.min.js
- FileSaver.min.js

If no web server is available, **tcsBU** can be saved into any local directory and opened directly in a browser using the `file://` protocol.

Workflow overview

The typical workflow in **tcsBU** follows four sequential steps:

> **Step 1:** Load a GML network file produced by TCS (*Load Data* button) → the haplotype network is drawn.

> **Step 2:** Load a group configuration file from the *Groups* tab → classification groups (e.g., geographic locations) with their colors and patterns are defined.

> **Step 3:** Load a haplotype configuration file from the *Haplotypes* tab → each sequence is assigned to a group; optionally, haplotype names (H1, H2, …) are assigned to nodes.

> **Step 4 (optional):** Load a trait configuration file from the *Traits* tab → a continuous variable (e.g., collection time or body size) is associated with each sequence and visualized as an inner ring gradient.

> **Important:** Always load or define groups (Step 2) *before* loading haplotype classifications (Step 3), so that group names can be matched correctly.

Usage

Step 1: Loading a GML file

Click the **Load Data** button (top-left of the toolbar) to open a file browser. Select the GML file produced by TCS — these files typically have the extension *.graph* and are found in the directory where the alignment used by TCS was stored. This action calls the internal `loadGraph` function, which parses the file, builds the internal node and edge lists, and starts the force-directed layout algorithm to display the raw haplotype network.

The GML (*Graph Modeling Language*) format is a plain-text format for network data. The first few lines of a typical TCS GML file look like this:

```
graph [
   directed 0
   node [
      data [
         Weight "outgroup weight=   0.01"
         Frequency "frequency=3
 Seq1
 Seq4
 Seq5
"
         Sequence "Sequence =
AAA
"
      ]
      id 0
      label "Seq1"
      graphics [
.
.
.
```

This file should not be modified unless strictly necessary. The parser relies on the formatting and absolute position of certain keywords, so any editing must preserve the original indentation.

Haplotype labels in TCS files

TCS does not restrict haplotype names beyond requiring standard ASCII characters. However, the *d3.js* library does not allow SVG element IDs to start with digits. Therefore, **tcsBU will automatically prepend an 'L' to every label that starts with a digit (0–9)**. This only matters when preparing haplotype configuration files (hapconf): any haplotype name that originally starts with a digit must also be written with a leading 'L' in the hapconf file. The safest practice is to avoid haplotype names that begin with a digit in the original alignment.

Step 2: Defining classification groups

Groups are managed in the **Groups** tab on the left panel. After loading a GML file, a 'Default' group (white, no pattern) is always present and cannot be modified or deleted. Any haplotypes not explicitly assigned to a group will belong to the Default group.

New groups can be added or deleted using the toolbar buttons within the Groups tab. Group names can be edited by double-clicking the name cell. **No two groups may share the same name**, as names are used as primary keys internally.

Each group (except Default) can be assigned a color and a fill pattern. Colors are specified in hexadecimal notation (e.g., `#AF0026`). Available pattern names are:

```
circles-1, circles-2,
lines-1, lines-2, lines-3, lines-4, lines-5, lines-6, lines-7, lines-8,
cross-1, cross-2
```

Click the **Load** button in the Groups tab toolbar to load a previously saved group configuration file (groupconf). This calls the internal `loadGroups` function. The Save button exports the current group list to a CSV file for later reuse. See Group configuration file format for details.

Step 3: Classifying haplotypes

After groups have been defined, click the **Load** button in the **Haplotypes** tab to load a haplotype configuration file (hapconf). This calls the internal `loadHaplotypes` function. The file assigns each sequence to a group and, optionally, gives a haplotype name to each node.

tcsBU supports two hapconf file formats, detected automatically by the number of columns:

Columns	Column 1	Column 2	Column 3	Effect
2	Sequence name	Group name	—	Assigns each sequence to a group. Node labels remain the first sequence name in each node.
3	Sequence name	Group name	Haplotype name (e.g., H1, H2)	Assigns sequences to groups *and* sets a custom haplotype label (e.g., H1) for each node. The **Show Hap Labels** toolbar button is enabled only when a 3-column file is loaded.

Sequence names must exactly match those in the original GML file (remember to prepend 'L' where names started with a digit). Group names must match those defined in the Groups tab. See Haplotype configuration file format for examples.

Step 4: Loading continuous trait data

A continuous trait (such as collection year, body size, or any other numeric variable) can be associated with sequences by clicking the **Load** button in the **Traits** tab. This calls the internal `loadTraits` function and reads a semicolon-delimited file with two columns: sequence name and numeric value. No header row is expected.

After loading, **tcsBU**:

- Determines the minimum and maximum trait values across all matched sequences.
- Maps each value to a color position in the selected gradient palette (Gray, Red, Green, or Blue) using a linear scale.
- Updates the `timeProportions` field of each node in the internal node list.
- Draws an **inner ring** inside each haplotype node. Each sector of the inner ring corresponds to one or more sequences sharing the same trait value, colored according to the gradient. The outer ring retains the group color/pattern information.

The gradient color palette can be changed at any time using the **Color** drop-down in the toolbar (options: Gray, Red, Green, Blue). A gradient legend bar showing the minimum and maximum trait values is displayed when the Legend is toggled on.

See Trait configuration file format for the expected file structure.

Display styles

The **Style** drop-down button in the toolbar controls how group and trait information are rendered on nodes. Three modes are available:

Style	Description
Dual-Trait	Default mode. If trait data has been loaded, the outer ring of each node shows group colors/patterns and the inner ring shows the continuous trait gradient. Both legends (group legend and gradient bar) are shown when the Legend is toggled on.
Category	Only group colors/patterns are shown; the inner trait ring is hidden even if trait data is loaded. Only the group legend is shown.
Continuous	Only the continuous trait gradient is shown; the group colors/patterns are hidden. Only the gradient legend bar is shown. Requires trait data to be loaded first.

> **Note:** Selecting *Dual-Trait* or *Continuous* style without first loading a trait file will produce a warning. Load the traitconf file before switching to these styles.

Force-directed layout algorithm

Once a GML file is loaded, the force-directed layout algorithm automatically starts and positions the nodes. The algorithm models haplogroups as electrically charged particles that repel each other, while edges act as springs that hold connected nodes together. Layout parameters can be adjusted in the **Advanced** panel (toolbar button):

- **Friction [0–1]:** velocity decay per iteration. A value of 1 is frictionless; 0 freezes all nodes.
- **Charge:** electrostatic charge of each node (scales with node size). Negative values cause repulsion (recommended). The more negative, the stronger the repulsion.
- **Gravity [0–1]:** a weak spring pulling each node toward the center of the canvas, preventing the network from drifting outside the visible area.

Link distance is controlled separately via the **Distance** toolbar button. Default values work well for most data sets; avoid adjusting them unless necessary.

Editing the network

Once a network is displayed, it can be edited interactively using the toolbar buttons:

Button	Function
Zoom In / Zoom Out	Scales the view in or out. The mouse wheel can also be used for zooming; clicking and dragging pans the view.
Delete Node (toggle)	When toggled on (button turns yellow), clicking any node deletes it from the network. Hovering over a node turns it red as a preview.
Delete Link (toggle)	When toggled on, clicking any edge (link) deletes it. Hovering over a link turns it red as a preview.
Outline (toggle)	Draws a thin black border between sectors of each pie-chart node, improving readability when many groups are displayed.
Show Hap Labels (toggle)	Displays the haplotype name (e.g., H1, H2) next to each haplotype node. Only available after a 3-column hapconf file has been loaded.
Legend (toggle)	Shows or hides the legend. The legend is draggable and can be repositioned anywhere on the canvas. Depending on the current display style, the legend shows a group color/pattern key and/or a gradient bar for the continuous trait.
Distance	Opens a panel to adjust the link distance parameter of the force-directed layout, stretching or compressing the network.
Advanced	Opens the advanced force-layout parameter panel (Friction, Charge, Gravity, line width).
Style	Selects the display style (Dual-Trait, Category, or Continuous). See Display styles.
Color	Selects the gradient color palette for the continuous trait: Gray, Red, Green, or Blue.

Individual nodes can also be dragged to any position by clicking and holding. Releasing a node will re-engage the force-directed simulation from that position.

Saving the network

Click the **Save SVG** button to export the entire network as a Scalable Vector Graphics (SVG) text file. All parts of the network — including any nodes currently outside the visible area — are included in the output. The legend is not scalable in SVG, so it is advisable to scale the network appropriately before saving. The resulting SVG file can be embedded directly in a modern word processor or further edited and exported using vector graphics software such as Inkscape or Adobe Illustrator®.

File formats

Group configuration file (groupconf)

A semicolon-delimited text file with no header row. Each line defines one group:

```
GroupName;#HexColor;PatternName
```

Color and pattern are optional. Examples:

```
Spain;#0af5d3;circles-1
Morocco South;#aea4b6
Morocco North;#54a1f2
Italy;#7684fa
Sardinia;#2425aa;lines-1
Portugal;#af0fff
France;#aaaa15
```

- Each line must have at least a group name.
- Colors must be in hexadecimal format (`#000000`–`#ffffff`); invalid colors are silently ignored.
- Pattern names must match one of the allowed values listed in Step 2; unknown pattern names are silently ignored.

Haplotype configuration file (hapconf)

A semicolon-delimited text file with no header row. Two formats are supported:

Two-column format (sequence name + group name):

```
L1543_IT-GIO;Italy
L1300_IT-SMR;Italy
L1860_MA-ZAO;Morocco
L2530_ES-IAL;Spain
L1821_ES-HER;Northern Spain
L1879_MA-RAS;Morocco
```

Three-column format (sequence name + group name + haplotype label):

```
L1543_IT-GIO;Italy;H1
L1300_IT-SMR;Italy;H1
L1860_MA-ZAO;Morocco;H2
L2530_ES-IAL;Spain;H3
L1821_ES-HER;Northern Spain;H4
L1879_MA-RAS;Morocco;H2
```

When a three-column file is loaded, sequences sharing the same haplotype label (column 3) are grouped into the same network node and that label is used as the node's display name. The **Show Hap Labels** toolbar button becomes enabled.

> **Reminder:** Sequence names must exactly match those in the GML file. If any name originally started with a digit, it will have been automatically prepended with 'L' by **tcsBU** — the same 'L' prefix must be used here. Group names must match those defined in the Groups tab.

Trait configuration file (traitconf)

A semicolon-delimited text file with no header row. Each line associates one sequence with a numeric trait value:

```
SequenceName;NumericValue
```

Example (collection year as trait):

```
L1543_IT-GIO;2005
L1300_IT-SMR;2008
L1860_MA-ZAO;2011
L2530_ES-IAL;2003
L1821_ES-HER;2015
L1879_MA-RAS;2009
```

- The numeric value can be an integer or decimal number.
- Sequences not present in the currently loaded network are silently ignored.
- A linear color scale maps the minimum value to one end of the gradient and the maximum value to the other.
- Within each node, the inner ring is subdivided into sectors proportional to the number of sequences at each distinct trait value.

References

- Clement M, Posada D, Crandall KA (2000) TCS: a computer program to estimate gene genealogies. *Molecular Ecology* 9, 1657–1659.
- Templeton AR, Crandall KA, Sing CF (1992) A cladistic analysis of phenotypic associations with haplotypes inferred from restriction endonuclease mapping and DNA sequence data. III. Cladogram estimation. *Genetics* 132, 619–633.
- Santos AM, Cabezas MP, Tavares AI, Xavier R, Branco M (2016) tcsBU: a tool to extend TCS network layout and visualization. *Bioinformatics* 32(4), 627–8. (doi: 10.1093/bioinformatics/btv636)